

Gene membership of the forty finalized, non-overlapping CCA subpaths. Subpaths ranked by pan-CCA selection intensity (P1, highest). Source gene set is the original definition each subpath was seeded from; gene sets were derived from either cancer gene-panel-restricted or exome-wide candidates. Genes within each subpath are listed by descending mutation count.

Subpath	Source gene set (MSigDB ID)	Cancer gene panel subpath	N genes	Member genes
P1	RalA downstream regulated genes (M39408)	Yes	3	KRAS, NRAS, RAC1
P2	Sterol regulatory element-binding proteins (SREBP) signaling (M39395)	Yes	4	PIK3CA, MTOR, AKT1, CDK8
P3	N-glycan trimming in the ER and Calnexin/Calreticulin cycle (M918)	No	25	MOGS, UGGT1, MARCHF6, MAN1B1, ENGASE, RNF103, AMFR, EDEM3, NGLY1, VCP, EDEM2, PRKCSH, UBXN1, PDIA3, CANX, RNF139, SYVN1, CALR, GANAB, RPS27A, EDEM1, TRIM13, UGGT2, DERL1, OS9
P4	Regucalcin in proximal tubule epithelial kidney cells (M39893)	Yes	7	SMAD4, BRAF, TGFBR1, CASP8, RAF1, SMAD2, MAPK1
P5	Ensembl 109 genes in cytogenetic band chr3p24 (M2406)	No	16	SLC4A7, KCNH8, SATB1, ZNF385D, DAZL, PLCL2, RARB, KAT2B, SGO1, TBC1D5, UBE2E2, NR1D2, AZI2, RBMS3, RFTN1, EOMES
P6	Glutathione metabolism (M1840)	No	22	GSS, GCLC, ANPEP, GPX5, GSTP1, PGD, TXNDC12, MGST1, LAP3, GSTA5, GGT7, ODC1, RRM2, IDH1, GSTK1, CHURC1, GPX4, G6PD, GSTA2, GCLM, RRM2B, RRM1
P7	Peroxisomal lipid metabolism (M2286)	No	22	CROT, ACBD5, HACL1, SLC25A17, SLC27A2, MLYCD, PECR, ACOX2, HSD17B4, EHHADH, ACOXL, AMACR, ABCD1, ACBD4, NUDT7, DECR2, NUDT19, CRAT, ALDH3A2, ACOX3, ACOT4, ACOX1

P8	Orc1 removal from chromatin (M4138)	No	38	MCM8, PSMD3, PSMA5, PSMA4, MCM6, MCM5, MCM7, MCM2, MCM3, CCNA1, ORC1, ORC4, ORC2, CCNA2, CDT1, PSME4, CDC6, PSMD1, PSMD12, PSMA7, PSMB7, ORC3, PSMD2, ORC6, PSMB2, PSMD9, PSMB3, PSME3, CDK2, ORC5, SKP2, PSMB5, PSMC5, PSMA3, CUL1, MCM4, PSMC3, RBX1
P9	Gastric cancer network 1 (M39347)	No	14	CENPF, CEP192, KIF15, KIF20B, LIN9, H4C16, ECT2, MYBL2, E2F7, TPX2, ACTL6A, RNF216, UBE2C, NUP107
P10	Influenza Infection (M4669)	No	30	CLTC, IPO5, POLR2B, POLR2A, KPNA5, GTF2F1, RPL3L, HSP90AA1, CPSF4, RPL9, RPL26L1, RPS27, RPL10L, DNAJC3, RPS9, GTF2F2, RPS2, KPNA4, ISG15, RPS3, RPL36A-HNRNPH2, TGFB1, RPS19, RPL23A, KPNA2, RPS15, RPL31, RPL10, HSPA1A, RPL6
P11	Interleukin-12 signaling (M27894)	No	21	TYK2, STAT4, HNRNPDL, IL12RB2, HNRNPA2B1, CAPZA1, IFNG, LMNB1, IL12B, P4HB, LCP1, IL12A, SOD2, CA1, ANXA2, HNRNPF, SERPINB2, MSN, TALDO1, ENSG00000251357, PDCD4
P12	Nuclear import of Rev protein (M27041)	No	23	TPR, NUP210, NUP98, NUP133, RANBP2, NDC1, NUP188, NUP214, NUP160, NUP155, NUP88, NUP54, NUP205, NUP153, SEH1L, SEC13, KPNB1, NUP37, NUP35, NUP93, NUP43, RAE1, NUP62
P13	Deadenylation-dependent mRNA decay (M839)	No	33	EDC4, EIF4G1, TUT4, CNOT1, SKIC3, CNOT10, EXOSC4, XRN1, NT5C3B, PAN2, TNKS1BP1, CNOT2, TUT7, SKIC8, CNOT11, LSM1, CNOT3, CNOT4, DCP2, EXOSC1, LSM5, DCP1B, EDC3, LSM4, EXOSC5, EXOSC9, EXOSC8, EXOSC7, PABPC1, EIF4A2, DCPS, PAN3, LSM3
P14	Gastric cancer network 2 (M39497)	Yes	4	TP53, CTNNB1, EGFR, MYC

P15	Beta5 beta6 beta7 and beta8 integrin cell surface interactions (M212)	No	14	FBN1, FN1, ITGB8, PLAUI, ITGA4, VCAM1, ITGB6, MADCAM1, ITGAV, VTN, CCN1, ITGB7, ITGB5, SDC1
P16	RNA Polymerase I Promoter Escape (M27691)	No	31	POLR1A, H3C3, TAF1C, TAF1B, UBTF, H3-3B, H2AC8, H3C11, H3C12, POLR1F, POLR1D, H4C8, H4C13, CCNH, TBP, H2BC1, GTF2H1, POLR1E, H3C7, H3C1, H3C8, TAF1D, H3C10, GTF2H3, H2BC8, H2BC6, H2BC4, H4C1, POLR1C, H4C5, RRN3
P17	Association of TriC/CCT with target proteins during biosynthesis (M3871)	No	25	ARFGEF2, LONP2, FBXO4, GBA1, WRAP53, XRN2, USP11, KIFC3, HDAC3, CCT2, CCT6B, FBXW5, STAT3, CCT6A, GAPDHS, FBXW10, NOP56, FBXW2, CCNE2, SPHK1, SIAH1, CCT5, CCT7, FBXO6, FBXL5
P18	Ensembl 109 genes in cytogenetic band chr4p14 (M7314)	No	20	TLR10, N4BP2, RBM47, KLB, KLHL5, RFC1, TBC1D1, C4orf19, TLR1, KLF3, LIAS, PGM2, NSUN7, ARAP2, UGDH, SMIM14, TLR6, RHOH, TMEM156, CHRNA9
P19	Aspirin ADME (M45014)	No	18	ALB, ABCC3, UGT3A1, BCHE, ABCC2, SLC16A1, ACSM5, CYP2C19, CES1, GLYAT, SLCO2B1, CES2, UGT1A5, UGT1A7, UGT2B4, UGT1A4, UGT3A2, BSG
P20	E3 ubiquitin ligases ubiquitinate target proteins (M27763)	No	20	CDC73, RNF40, LEO1, WAC, CTR9, HLTF, PAF1, SHPRH, PEX5, HLA-A, RNF152, RRAGA, PCNA, UBE2D3, RTF1, RNF20, RAD18, PEX10, UBE2V2, PEX13
P21	TGF-beta receptor signaling (M286)	No	27	SPTBN1, SOS1, ITCH, SMURF1, RNF111, WWP1, DAB2, ZFYVE16, TGFB3, ZFYVE9, SKIL, CAMK2A, BAMBI, YWHAE, SHC1, SMURF2, TGFB2, TAB1, TAB2, YAP1, PPP2R2A, SMAD7, PML, PPP1R15A, CCN2, RPS6KB1, PARD6A
P22	Ensembl 109 genes in cytogenetic band chr4q26 (M6)	No	13	SYNPO2, USP53, METTL14, NDST4, NDST3, TRAM1L1, PRSS12, UGT8, PDE5A, SEC24D, MYOZ2, CAMK2D, ARSJ

P23	Valine, leucine and isoleucine degradation (M11835)	No	24	AOX1, PCCB, ALDH2, MMUT, ALDH1B1, HADHA, PCCA, DLD, ACADS, HMGCL, ALDH9A1, HADH, IVD, BCAT2, ACADSB, ACADM, ACAD8, DBT, ECHS1, BCKDHA, HMGCS1, AUH, OXCT2, ALDH7A1
P24	Retrograde transport at the Trans-Golgi-Network (M27653)	No	29	IGF2R, GOLGA4, VPS53, RHOBTB3, COG2, COG7, GOLGA1, GCC1, COG3, STX6, TMF1, RIC1, PLIN3, COG6, VPS54, COG4, COG5, NSF, ARFIP2, RABEPK, COG8, NAA35, STX10, NAPA, VPS51, NAA30, GCC2, SCOC, VAMP4
P25	Ensembl 109 genes in cytogenetic band chr3p21 (M2833)	Yes	4	BAP1, PBRM1, SETD2, RHOA
P26	22q11.2 copy number variation syndrome (M39787)	Yes	7	FGFR2, NCOR1, FGFR1, PRKN, XPO1, CRKL, CUL3
P27	ErbB2/ErbB3 signaling events (M175)	Yes	13	ERBB3, ERBB2, PIK3R1, MAP2K1, JAK2, PIK3CB, PIK3R3, NF2, PIK3CD, SRC, JUN, PIK3R2, MAP2K2
P28	Meiotic recombination (M1011)	Yes	13	ATM, BRCA2, BRCA1, RAD51C, NBN, MRE11, MLH1, BLM, RAD50, RAD51, CDK4, H3C2, H2BC5
P29	Metabolism of RNA (M16843)	Yes	7	SF3B1, RBM10, ERCC3, ERCC2, U2AF1, PPP2R1A, DIS3
P30	Activation of anterior HOX genes in hindbrain development during early embryogenesis (M29689)	Yes	9	KMT2C, EP300, CREBBP, KDM6A, RARA, EZH2, CTCF, SUZ12, EED
P31	VEGFA-VEGFR2 signaling (M39729)	Yes	15	IDH2, KDR, MAP2K4, ABL1, FLT1, ERG, NOTCH4, CBL, RICTOR, BRD4, INPP4B, MDM2, BCL2L1, CCND1, PAK1
P32	Chromosomal and microsatellite instability in colorectal cancer (M39758)	Yes	9	TGFBR2, ARAF, AXIN1, MSH6, MSH2, AXIN2, SMAD3, CDKN1A, REL
P33	Transcriptional Regulation by TP53 (M27267)	Yes	21	ATR, PTEN, RPTOR, BARD1, TSC1, PRDM1, CDK12, TSC2, DAXX, CDKN1B, BCL6, MDC1, PLK2, PMS2, CCNE1, BRIP1, CHEK2, AURKB, MDM4, RAD51D, CHEK1

P34	Apoptosis (M15303)	Yes	10	APC, CDKN2A, TP63, CDH1, H1-2, AKT3, BCL2L11, XIAP, AKT2, BBC3
P35	Ensembl 109 genes in cytogenetic band chr1p36 (M19292)	Yes	3	ARID1A, SPEN, SDHB
P36	EGFR tyrosine kinase inhibitor resistance (M39839)	Yes	10	NF1, IGF1R, PDGFRB, FGFR3, MET, AXL, JAK1, PDGFRA, HGF, RPS6KB2
P37	Pathways affected in adenoid cystic carcinoma (M39682)	Yes	9	FBXW7, NSD1, NOTCH1, ATRX, BCOR, FGFR4, ARID5B, MYCN, MAX
P38	RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known (M27797)	Yes	5	ARID2, ARID1B, SMARCA4, SMARCB1, SMARCD1
P39	Infectious disease (M27548)	Yes	19	GNAS, RB1, JAK3, NFE2L2, RUNX1, KEAP1, BTK, IFNGR1, PDCD1, PIK3C3, B2M, SYK, VHL, PARP1, PRKAR1A, IKBKE, IL10, YES1, TMPRSS2
P40	MAPK family signaling cascades (M27565)	Yes	10	RASA1, ERBB4, IRS2, RET, FLT3, KIT, IRS1, FGF3, FGF19, FGF4
